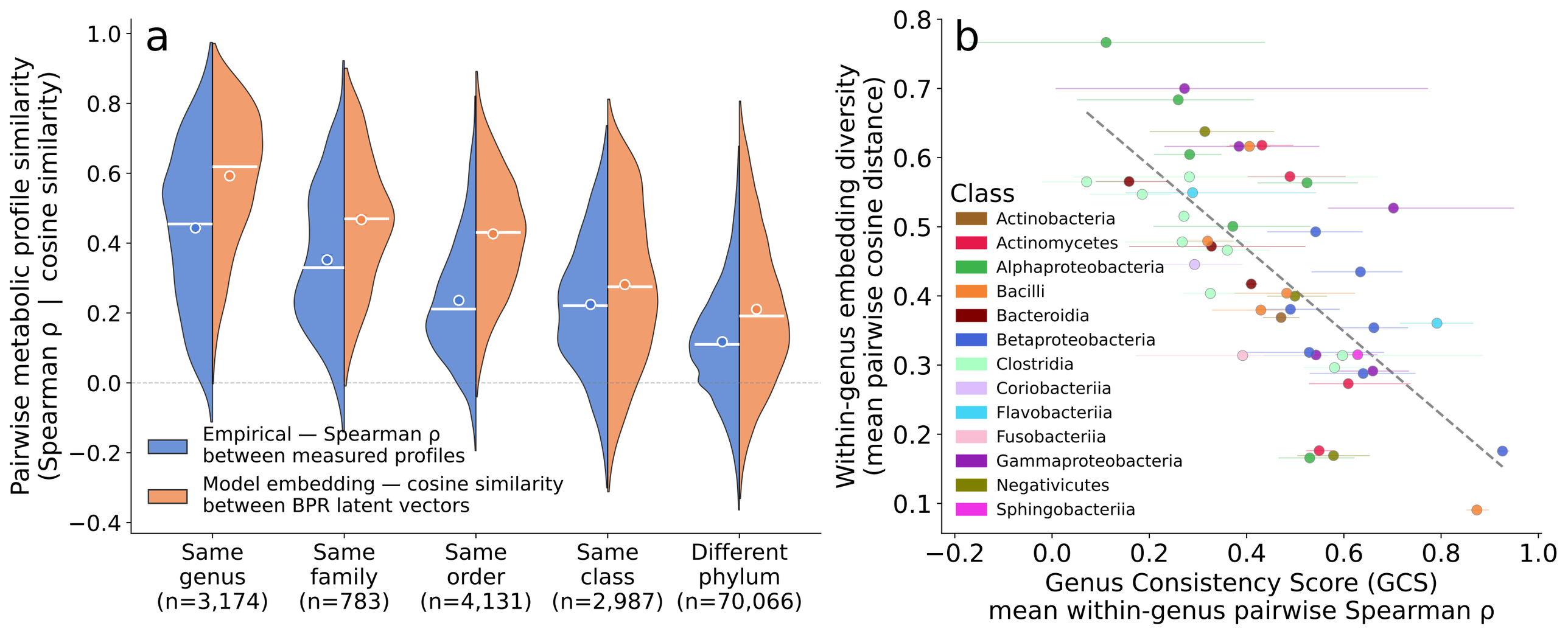

a
Pairwise metabolic profile similarity
(Spearman ρ | cosine similarity)
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
Empirical — Spearman ρ
between measured profiles
Model embedding — cosine similarity
between BPR latent vectors
Same
genus
(n=3,174)
Same
family
(n=783)
Same
order
(n=4,131)
Same
class
(n=2,987)
Different
phylum
(n=70,066)
b
Within-genus embedding diversity
(mean pairwise cosine distance)
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
Class
Actinobacteria
Actinomycetes
Alphaproteobacteria
Bacilli
Bacteroidia
Betaproteobacteria
Clostridia
Coriobacteriia
Flavobacteriia
Fusobacteriia
Gammaproteobacteria
Negativicutes
Sphingobacteriia
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Genus Consistency Score (GCS)
mean within-genus pairwise Spearman ρ